

Arunabha Majumder

Mechanical Engineer — Lab Hardware, Rapid Prototyping & Systems Integration (SolidWorks)

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October 7, 2026

Hardware Pre-Development Team
QuantumDiamonds GmbH
Munich, Germany

Application: Mechanical Engineer (Optomechanics & Quantum Sensing)

Dear QuantumDiamonds Team,

I am applying for the Mechanical Engineer (Optomechanics & Quantum Sensing) position in your Hardware Pre-Development Team. I am a mechanical engineer finishing my PhD at Aalborg University, and I enjoy exactly what you describe: turning the requirements of scientists and engineers into pragmatic mechanical designs, and building, testing and rebuilding them myself until they work.

In my PhD, I designed and built lab hardware from concept to working system. I created the assemblies in SolidWorks, chose the right process for each part, from 3D printing on our in-house printers to machined and off-the-shelf components, and integrated actuators, a motor, encoder, force sensors, DAQ and embedded control into one setup. I used FEA to keep load-carrying structures stiff and light. The resulting design cut energy consumption by more than 25% and won the Best Paper Award at IFToMM ISRM 2026.

I think naturally in geometry and systems. On a 5-bar planar parallel robot, I modelled the kinematics and estimated the end-effector force from the robot's actuators through its Jacobian, validated against a force sensor to below 10% RMS error, and used the method in lab measurements. During my Master's at CSIR-CMERI, I built a complete pneumatic test setup from raw materials.

I have not yet worked with optics, RF or magnetic environments, tolerance budgets or non-magnetic materials, and I would be glad to learn them from your physicists and engineers. I am curious about the physics behind the Quantum Diamond Microscope, and I would much rather test a quick prototype than debate a design on paper. I am open to relocating to Munich.

I would welcome the opportunity to discuss how I can contribute to your demonstrators. References are available upon request.

Kind regards,
Arunabha Majumder